

CS345: Design and Analysis of Algorithms

Assignment 1

Due Date: 31st August

Total Number of Pages: 1

Total Points 40

Instructions-

1. For submission typeset the solution to each problem and compile them in a single pdf file. Hand-written solutions will not be accepted. You can use L^AT_EX or Word for typesetting.
 2. Start each problem from a new page. Write down your Name, Roll number and problem number clearly for each problem.
 3. For each question, give the pseudo-code of the algorithm with a clear description of the algorithm. Unclear description will receive less marks. Less optimal solutions will receive only partial marks.
 4. Assume that sorting would have $O(n \log n)$ complexity.
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1 Problems that need to be submitted

Question 1. (10 points) Given two positive numbers n and m , your task is to design a $O(n \log m)$ time complexity algorithm to compute the n th root of m (i.e. $\sqrt[n]{m}$) using a **Divide and Conquer** approach. Alternatively, your algorithm should find the value of x such that x^n is equal to m . Assume that n th root of m is a natural number.

Question 2. (10 points) Given a polygon (not necessarily convex) P on n points, and a point α , design an $O(n)$ time algorithm to check if α is contained in P or not.

Question 3. (10 points) A photocopy shop has a single large machine. Each morning the shop receives a set of jobs from customers. The shopkeeper wants to do the jobs on the single photocopying machine in an order that keeps their customers happiest. Customer i 's job will take t_i time to complete. Given a schedule (ordering of the jobs), let C_i denote the finishing time of job i . For example, if job i is the first to be done, we would have $C_i = t_i$; and if job j is done right after job i , we would have $C_j = C_i + t_j$. Each customer has a given weight w_i that represents his or her importance to the business. The happiness of customer i is expected to be dependent on the finishing time of i 's job. So the company decides that they want to order the jobs to minimize the weighted sum of the completion time, $\sum_{i=1}^n w_i C_i$.

Design an efficient algorithm to solve this problem. That is, you are given a set of n jobs: job i has a processing time t_i and a weight w_i . You want to order the jobs so as to minimize the weighted sum of the completion time, $\sum_{i=1}^n w_i C_i$.

Question 4. (10 points) You are given a directed acyclic graph $G = (V, E)$ in which each node $u \in V$ has an associated price, denoted by $price(u)$, which is a positive integer. The cost of a node u , denoted by $cost(u)$, is defined to be the price of the cheapest node reachable from u (including u itself). Design an algorithm that computes $cost(u)$ for all $u \in V$.